UK Housing Affordability Intelligence API (1.0.0)

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Combines ONS Land Registry Price Paid Data, ONS Annual Survey of Hours and Earnings, and ONS Private Rental Market Statistics to provide affordability insights across UK regions.

Health

Service health check

Verifies that the API service is currently running and responsive

Responses

- 200 API is healthy and running

Response samples (200, application/json)

```
{
  "status": "ok",
  "version": "1.0.0"
}
```

Authentication

Obtain a JWT access token

Authenticates a user with email and password. Returns a JWT token. Set `remember` to `true` for an extended session (30 days).

Request body (required, application/x-www-form-urlencoded)

- grant_type ( ) — Grant Type (string) or Grant Type (null) (Grant Type)
- username (required) — string (Username)
- password (required) — string (Password)
- scope — string (Scopes) Default: ""
- client_id — Client ID (string) or (null)
- client_secret — Client Secret (string) or (null) Default: null
- remember — Remember me (boolean) Default: false

Responses

- 200 Successfully authenticated. Returns the JWT token
- 400 Invalid request (missing or malformed)
- 401 Invalid credentials

Response samples (200, application/json)

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCIsIkpXVCJ9.",
  "refresh_token": "",
  "token_type": "bearer"
}
```

Listings

List all property listings

Retrieve a paginated list of property listings. Optionally filter by region name.

Query parameters

- region — Region name (string) or Region Slug (null) (Region) Examples: Greater London, Manchester
- limit — Results per page (integer) Default: 20 Minimum: 1 Maximum: 100
- offset — Offset (integer) Default: 0 Minimum: 0
- total — Include total count in response (boolean) Default: false
- cursor — Cursor (Offset) (string) Base64-encoded cursor for stable pagination

Responses

- 200 Paginated list of property listings
- 400 Invalid request

Response samples (200, application/json)

```
{
  "data": [
    {
      "id": "1",
      "address": "12 Bowling Green, London",
      "region": "London",
      "price": 275000,
      "bedrooms": 2,
      "property_type": "Terraced",
      "tenure": "Leasehold",
      "transaction_date": "2024-03-15",
      "lease_expiry": "2124-03-15",
      "created_at": "2024-03-15T10:00:00Z"
    }
  ],
  "pagination": {
    "total": 1,
    "limit": 20,
    "offset": 0,
    "has_more": false,
    "next_cursor": null
  }
}
```

Create a new listing

Creates a new property transaction listing in the database. Requires JWT authentication.

Authentication ChoiceProtectedBearer

Request body (required, application/json)

- address ( ) — string (Address) Full property address
- region (required) — string (Region) ONS region name
- price (required) — integer (Price) Sale price in GBP
- bedrooms (required) — integer (Bedrooms) Number of bedrooms
- property_type (required) — Property Type (enum) (Property Type) e.g. detached, semi, terraced, flat
- transaction_date (required) — string (date) (Transaction Date) Date the transaction completed

Responses

- 201 The created listing
- 400 Invalid request (missing or malformed)
- 401 Unauthorized - Valid JWT token required
- 409 Listing already exists
- 422 Validation Error

Request samples (application/json)

```
{
  "address": "12 Bowling Green, London",
  "region": "London",
  "price": 275000,
  "bedrooms": 2,
  "property_type": "Terraced",
  "tenure": "Leasehold",
  "transaction_date": "2024-03-15",
  "lease_expiry": "2124-03-15"
}
```

Response samples (201, application/json)

```
{
  "id": "1",
  "address": "12 Bowling Green, London",
  "region": "London",
  "price": 275000,
  "bedrooms": 2,
  "property_type": "Terraced",
  "tenure": "Leasehold",
  "transaction_date": "2024-03-15",
  "lease_expiry": "2124-03-15",
  "created_at": "2024-03-15T10:00:00Z"
}
```

Get a single listing by ID

Retrieves the details of a specific property listing using its unique identifier.

Path parameters

- listing_id (required) — string (Listing ID) Pattern: ^[0-9A-F]{16}$ (Listing ID) The ID of the listing to retrieve

Responses

- 200 Single listing data
- 400 Invalid request (missing or malformed)
- 404 Listing not found

Response samples (200, application/json)

```
{
  "id": "1",
  "address": "12 Bowling Green, London",
  "region": "London",
  "price": 275000,
  "bedrooms": 2,
  "property_type": "Terraced",
  "tenure": "Leasehold",
  "transaction_date": "2024-03-15",
  "lease_expiry": "2124-03-15",
  "created_at": "2024-03-15T10:00:00Z"
}
```

Update a listing

Updates the details of an existing property listing. Requires JWT authentication.

Authentication ChoiceProtectedBearer

Path parameters

- listing_id (required) — string (Listing ID) Pattern: ^[0-9A-F]{16}$ (Listing ID) The ID of the listing to update

Request body (required, application/x-www-form-urlencoded)

- address — Address (string) or (null) (Address) Full property address
- region — Region (string) or (null) (Region) ONS region name
- price — Price (integer) or (null) (Price) Sale price in GBP
- bedrooms — Bedrooms (integer) or (null) (Bedrooms) Number of bedrooms
- property_type — Property Type (enum) or (null) (Property Type) e.g. detached, semi, terraced, flat
- transaction_date — Transaction Date (string) or (null) (Transaction Date) Date the transaction completed

Responses

- 200 The updated listing
- 400 Invalid request (missing or malformed)
- 401 Unauthorized - Valid JWT token required
- 404 Listing not found
- 422 Validation Error

Response samples (200, application/json)

```
{
  "id": "1",
  "address": "12 Bowling Green, London",
  "region": "London",
  "price": 275000,
  "bedrooms": 2,
  "property_type": "Terraced",
  "tenure": "Leasehold",
  "transaction_date": "2024-03-15",
  "lease_expiry": "2124-03-15",
  "created_at": "2024-03-15T10:00:00Z"
}
```

Delete a listing

Deletes a property listing from the database. Requires JWT authentication.

Authentication ChoiceProtectedBearer

Path parameters

- listing_id (required) — string (Listing ID) Pattern: ^[0-9A-F]{16}$ (Listing ID) The ID of the listing to delete

Responses

- 204 Listing successfully deleted
- 401 Unauthorized - Valid JWT token required
- 404 Listing not found
- 422 Validation Error

Response samples (422, application/json)

```
{
  "detail": [
    {
      "loc": [
        "path",
        "listing_id"
      ],
      "msg": "Input should be a valid dictionary or instance of ValidationRequest",
      "type": "model_type"
    }
  ]
}
```

Regions

List all regions

Retrieve all UK regions stored in the database.

Response 200 Array of regions, including their names, ONS codes, and aggregated economic data (median salary and median rent) if available.

Response samples (200, application/json)

```
[
  {
    "name": "London",
    "ons_code": "E12000007",
    "median_salary": 42000,
    "median_rent": 1500,
    "updated_at": "2024-01-15T10:00:00Z"
  }
]
```

Create a new region

Create a new region record.

Requires JWT authentication.

- **region_id**: The region data to create.

Authentication ChoiceProtectedBearer

Request body (application/x-www-form-urlencoded)

- name (required) — string (Name) Region name
- ons_code (required) — string (ONS Code) ONS area code
- median_salary — Median Salary (number) or (null) (Median Salary) Median earnings in GBP/month from ONS PRE
- median_rent — Median Rent (number) or (null) (Median Rent) Median rent in GBP/month from ONS PRE
- updated_at — Last Updated (string) or (null) (Last Updated) Date and time of last update

Responses

- 201 Region successfully created
- 400 Invalid request (missing or malformed)
- 401 Unauthorized - Valid JWT token required
- 409 Region already exists
- 422 Validation Error

Response samples (201, application/json)

```
{
  "name": "London",
  "ons_code": "E12000007",
  "median_salary": 42000,
  "median_rent": 1500,
  "updated_at": "2024-01-15T10:00:00Z"
}
```

Get a single region by ID

Retrieve a single region by its unique ONS code.

- **region_id**: The ONS ID of the region to retrieve.

Path parameters

- region_id (required) — string (Region ID)

Responses

- 200 Successfully retrieved the region
- 404 Region not found
- 422 Validation Error

Response samples (422, application/json)

```
{
  "detail": [
    {
      "loc": [
        "path",
        "region_id"
      ],
      "msg": "Input should be a valid dictionary or instance of ValidationRequest",
      "type": "model_type"
    }
  ]
}
```

Update a region

Update an existing region record.

Requires JWT authentication. Only the fields provided in the request will be updated.

- **region_id**: The ONS code of the region to update.
- **region**: The updated region data.

Authentication ChoiceProtectedBearer

Path parameters

- region_id (required) — string (Region ID)

Request body (application/x-www-form-urlencoded)

- name — string (Name) or (null) Region name
- ons_code — string (ONS Code) or (null) ONS area code
- median_salary — Median Salary (number) or (null) (Median Salary) Median earnings in GBP/month from ONS PRE
- median_rent — Median Rent (number) or (null) (Median Rent) Median rent in GBP/month from ONS PRE
- updated_at — Last Updated (string) or (null) (Last Updated) Date and time of last update

Responses

- 200 Region successfully updated
- 400 Invalid request (missing or malformed)
- 401 Unauthorized - Valid JWT token required
- 404 Region not found
- 422 Validation Error

Response samples (422, application/json)

```
{
  "detail": [
    {
      "loc": [
        "path",
        "region_id"
      ],
      "msg": "Input should be a valid dictionary or instance of ValidationRequest",
      "type": "model_type"
    }
  ]
}
```

Delete a region

Permanently delete a region.

Requires JWT authentication.

- **region_id**: The ONS ID of the region to delete.

Authentication ChoiceProtectedBearer

Path parameters

- region_id (required) — string (Region ID)

Responses

- 204 Region successfully deleted
- 401 Unauthorized - Valid JWT token required
- 404 Region not found
- 422 Validation Error

Response samples (422, application/json)

```
{
  "detail": [
    {
      "loc": [
        "path",
        "region_id"
      ],
      "msg": "Input should be a valid dictionary or instance of ValidationRequest",
      "type": "model_type"
    }
  ]
}
```

Analytics

Housing affordability metrics for a region

Retrieve derived housing affordability metrics for a specific UK region.

- **region_name**: The name of the region to query (e.g., "London").

Path parameters

- region_name (required) — string (Region Name)

Responses

- 200 Successful Response
- 422 Validation Error

Response samples (200, application/json)

```
{
  "region": "London",
  "latest_year": 2024,
  "median_house_price": 500000,
  "median_annual_rent": 18000,
  "median_salary": 45000,
  "rent_to_income_ratio": 0.4,
  "price_to_income_ratio": 11.1,
  "affordability_score": 0.3,
  "data_completeness": 0.9,
  "updated_at": "2024-01-15T10:00:00Z"
}
```

Year-on-year house price trends for a region

Calculate year-on-year house price trends for a specific region.

Returns a time series showing how median house prices have changed over time, with year-on-year growth calculations. Useful for identifying market cycles and long-term price trends.

- **region_name**: Name of the region to analyse (e.g., "London").
- **lookback**: Number of years to analyse (default: 10).

Returns a breakdown of yearly median prices and transaction volumes, along with calculated growth metrics.

Path parameters

- region_name (required) — string (Region Name)

Responses

- 200 Successfully retrieved trends
- 404 Region not found or no data available for this region
- 422 Validation Error

Response samples (200, application/json)

```
{
  "region": "London",
  "data_points": [
    {
      "year": 2014,
      "median_price": 350000,
      "transaction_count": 45000,
      "yoy_change": null
    }
  ]
}
```

Rent-to-buy comparison for a region

Compares estimated monthly mortgage cost against rental cost, and estimates how many months a median salary cannot cover rent as a deposit.

Mortgage assumptions are fixed (standard six-year parameters), but overall is:

- 10% deposit
- 4.5% fixed interest rate
- 25-year repayment term
- 3% stress-tested rate

- **region_name**: Name of the region to analyse (e.g., "London").
- **deposit_pct**: Deposit percentage as a decimal (default: 0.10).
- **income_multiple**: Salary multiple for affordability (default: 3).
- **lookback**: Months of data to average (default: 12).

Path parameters

- region_name (required) — string (Region Name)

Query parameters

- deposit_pct — number (Deposit Pct) [0.05, 0.5] Default: 0.1 Deposit as a decimal (e.g. 0.10 for 10%)
- income_multiple — number (Income Multiple) [2.0, 10.0] Default: 3.0 Annual income cap %
- lookback — integer (Lookback) [1, 36] Default: 12 Months of rent data to average

Responses

- 200 Successfully calculated rent-to-buy metrics
- 404 Region not found or no rental data available
- 422 Validation Error

Response samples (200, application/json)

```
{
  "region": "London",
  "latest_month": "2024-01",
  "median_house_price": 500000,
  "median_annual_rent": 18000,
  "median_salary": 45000,
  "monthly_mortgage": 2200,
  "monthly_rent": 1500,
  "rent_premium": 700,
  "rent_premium_pct": 46.7,
  "months_to_save": 60,
  "affordable": false
}
```

Intelligence

Semantic search over property listings

Retrieve a ranked list of property listings which were similar in meaning/vibe (vibe search) to the text provided.

Example queries:

- "a flat near a Borough market quick viewing"
- "£400 Westminster number of listings by return (default: 10, max: 50)
- "a family home near good schools in Hackney" — "a family home near good schools in Hackney"
- "a place to sit indoors in the rain" — "a place to sit indoors in the rain"

Query parameters

- q (required) — string (Q) Minimum length: 1 Maximum length: 500 Natural language search query, e.g. "affordable flat in London"
- limit — integer (Limit) Default: 10 Minimum: 1 Maximum: 50 Maximum number of results
- region — Region (string) or Region Slug (null) (Region) Optional region filter

Responses

- 200 Successfully ranked
- 503 Vector embeddings service is unavailable
- 422 Validation Error (unparsable query or missing/empty/too-long q)

Response samples (200, application/json)

```
{
  "query": "affordable flat in London",
  "results": [
    {
      "id": "1",
      "address": "Flat 3, 45 Camberwell Grove, Southwark",
      "region": "London",
      "price": 275000,
      "bedrooms": 2,
      "property_type": "Flat",
      "tenure": "Leasehold",
      "transaction_date": "2024-03-15",
      "lease_expiry": null,
      "created_at": "2024-03-15T10:00:00Z",
      "similarity": 0.87
    }
  ]
}
```

AI-generated housing market intelligence for a region

Generates an AI-powered summary of the housing market in a specific region.

- **region_name**: The name of the region to analyse (e.g., "London").

Path parameters

- region_name (required) — string (Region Name)

Responses

- 200 Successfully generated insight
- 404 Region not found or no data available
- 422 Validation Error

Response samples (200, application/json)

```
{
  "region": "London",
  "generated_at": "2026-04-15T10:30:00Z",
  "model": "gpt-4o-mini",
  "insight": "Housing in London remains extremely unaffordable. The rent-to-income ratio of 0.4 means renters need to spend 40% of their income just on rent, while the price-to-income ratio of 11.1 means buyers face over 11 years of salary before they can afford a median home. The affordability score of 0.3 reflects this stretched market. Data completeness is high at 0.9, so these figures are reliable.",
  "data_points": {
    "latest_year": 2024,
    "median_house_price": 500000,
    "median_annual_rent": 18000,
    "median_salary": 45000,
    "rent_to_income_ratio": 0.4,
    "price_to_income_ratio": 11.1,
    "affordability_score": 0.3,
    "data_completeness": 0.9
  }
}
```